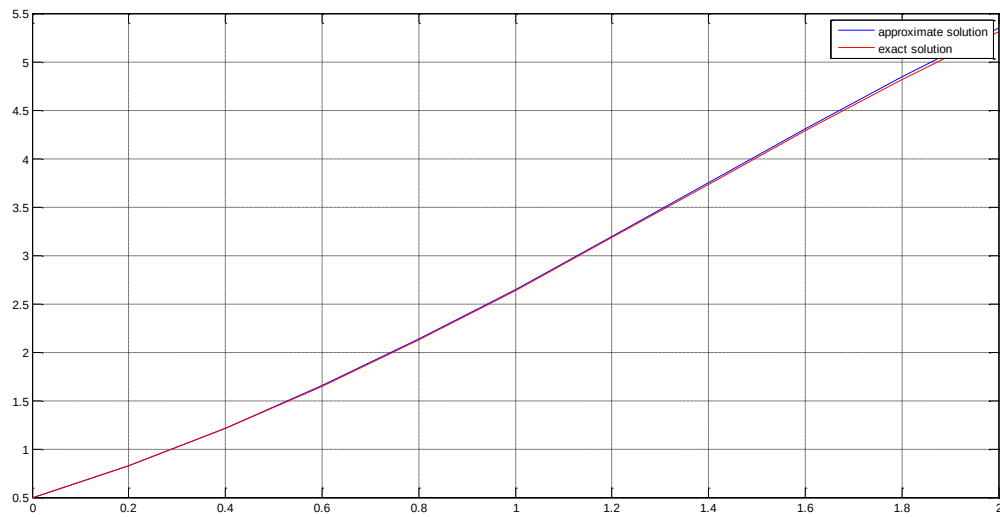


Grand Assignment File

If the first 4 digits of your student id are 1902 (for example bc190206641, mc190206623) then:

Solution:

```
clc
clearall
closeall
f=@(t,y)y-t.^2+1;
g=@(t,y)y-t.^2+1-2.*t;
n=10;
t0=0;
tn=2;
y0=0.5;
h=(tn-t0)/n;
t(1)=t0;
y(1)=y0;
for k=1:n
    t(k+1)=t(k)+h;
    a=f(t(k),y(k))*h;
    b=(h^2/2)*g(t(k),y(k));
    y(k+1)=y(k)+a+b;
end
tvalue=transpose(t)
yvalue=transpose(y)
plot(tvalue,yvalue,'b')
holdon
sol=dsolve('Dy=y-t^2+1','y(0)=0.5','t')
exact=eval(vectorize(sol))
plot(tvalue,exact,'r')
gridon
legend('approximate solution','exact solution')
```



If the first 4 digits of your student id are 1102 , 1702 , 1704 (for example bc110206641, mc170206623, bc170206623, mc170406623, bc170406623) then:

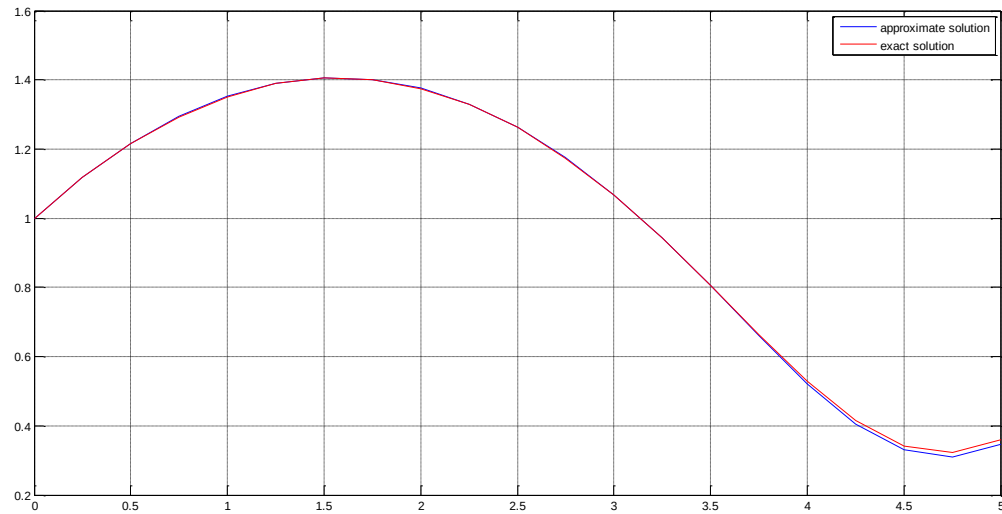
Solution:

```
clc
clearall
closeall
f=@(t,y)cos(t)/(1+y.^2);
g=@(t,y)-2.*y*cos(t)^2/(1+y.^2)^3-sin(t)/(1+y.^2);
n=20;
t0=0;
tn=5;
y0=1;
h=(tn-t0)/n;
t(1)=t0;
y(1)=y0;
for k=1:n
    t(k+1)=t(k)+h;
    a=f(t(k),y(k))*h;
    b=(h^2/2)*g(t(k),y(k));
    y(k+1)=y(k)+a+b;
end
tvalue=transpose(t)
yvalue=transpose(y)
plot(tvalue,yvalue,'b')
holdon
sol=dsolve('Dy=cos(t)/(1+y^2)','y(0)=1','t')
exact=eval(vectorize(sol))
```

```

plot(tvalue,exact,'r')
gridon
legend('approximate solution','exact solution')

```



If the first 4 digits of your student id are 1802 or 1804 (for example bc180405531, mc180405531, bc180203155, mc180203151) then:

Solution:

```

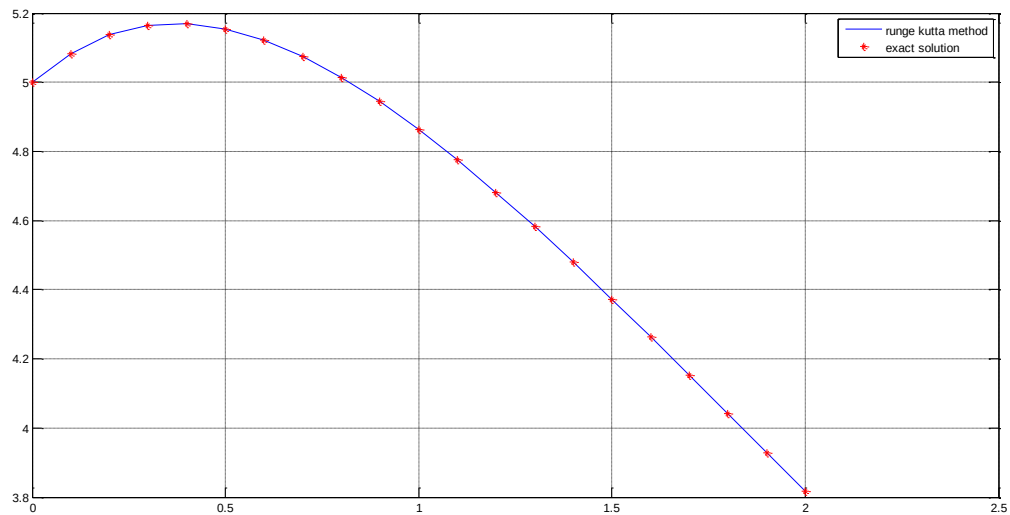
clc
clearall
closeall
f=@(t,y) 3.*exp(-t)-0.4.*y;
n=20;
t0=0;
tn=2;
y0=5;
h=(tn-t0)/n;
t(1)=t0;
y(1)=y0;
for i=1:n
    t(i+1)=t(i)+h;
    k1=f(t(i),y(i));
    k2=f(t(i)+h/2,y(i)+h*k1/2);
    k3=f(t(i)+h/2,y(i)+h*k2/2);
    k4=f(t(i)+h,y(i)+h*k3);
    y(i+1)=y(i)+1/6*h*(k1+2*k2+2*k3+k4);
end

```

```

tvalue=transpose(t)
yvalue=transpose(y)
plot(tvalue,yvalue,'b')
holdon
sol=dsolve('Dy=3*exp(-t)-0.4*y','y(0)=5','t')
exact=eval(vectorize(sol))
plot(tvalue,exact,'*r')
gridon
legend('rungekuttamethod','exact solution')

```



If the first 4 digits of your student id are 1904 , 2002 , 1004 , 1202 , 1304 , 1604 or any other ID not mentioned above then:

Solution:

```

clc
clearall
closeall
f=@(t,y) 2.*y+t.^2;
n=20;
t0=0;
tn=2;
y0=1;
h=(tn-t0)/n;
t(1)=t0;
y(1)=y0;
for i=1:n
    t(i+1)=t(i)+h;

```

```

    k1=f(t(i),y(i));
    k2=f(t(i)+h/2,y(i)+h*k1/2);
    k3=f(t(i)+h/2,y(i)+h*k2/2);
    k4=f(t(i)+h,y(i)+h*k3);
    y(i+1)=y(i)+1/6*h*(k1+2*k2+2*k3+k4);
end
tvalue=transpose(t)
yvalue=transpose(y)
plot(tvalue,yvalue,'b')
hold on
sol=dsolve('Dy=2*y+t^2','y(0)=1','t')
exact=eval(vectorize(sol))
plot(tvalue,exact,'*r')
grid on
legend('runge kutta method','exact solution')

```

